Breast Cancer Detection

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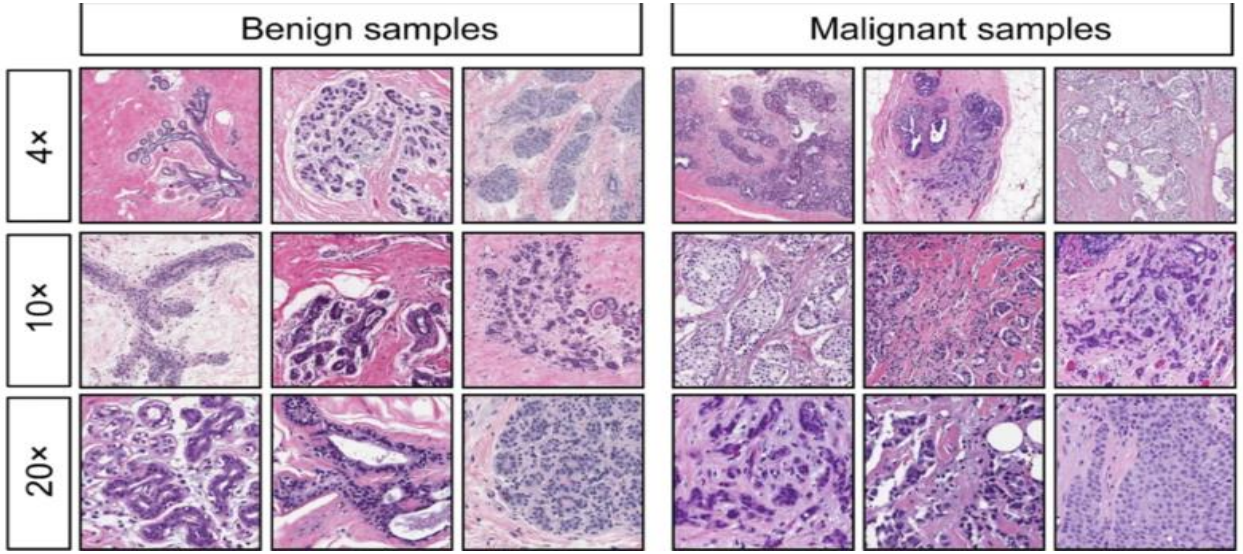
Problem Incentive

- Breast cancer is the second leading cause of cancer death
- Early detection improves survival
- Data analytics for early diagnosis which "99%–100% 5-year relative survival rate for localized (Stage I) cancer compared to roughly 31% for distant (metastatic) disease." (Shockney, 2023)
- Our research goal is to determine the classification of malignant or benign

Dataset

- Breast Cancer Wisconsin (Diagnostic) Dataset.
- Computed from a digitized image of a fine needle aspirate biopsy.
- 569 Observations
- 30 numeric features
- Features include radius, texture, perimeter and area.



Dataset Cleaning

- Classification: Malignant vs. Benign (in the code base 1 and 0 respectively).
- Dropped an unnamed column with only NaN datatypes

```
if "Unnamed: 32" in df.columns:
    df = df.drop(columns=["Unnamed: 32"])
```

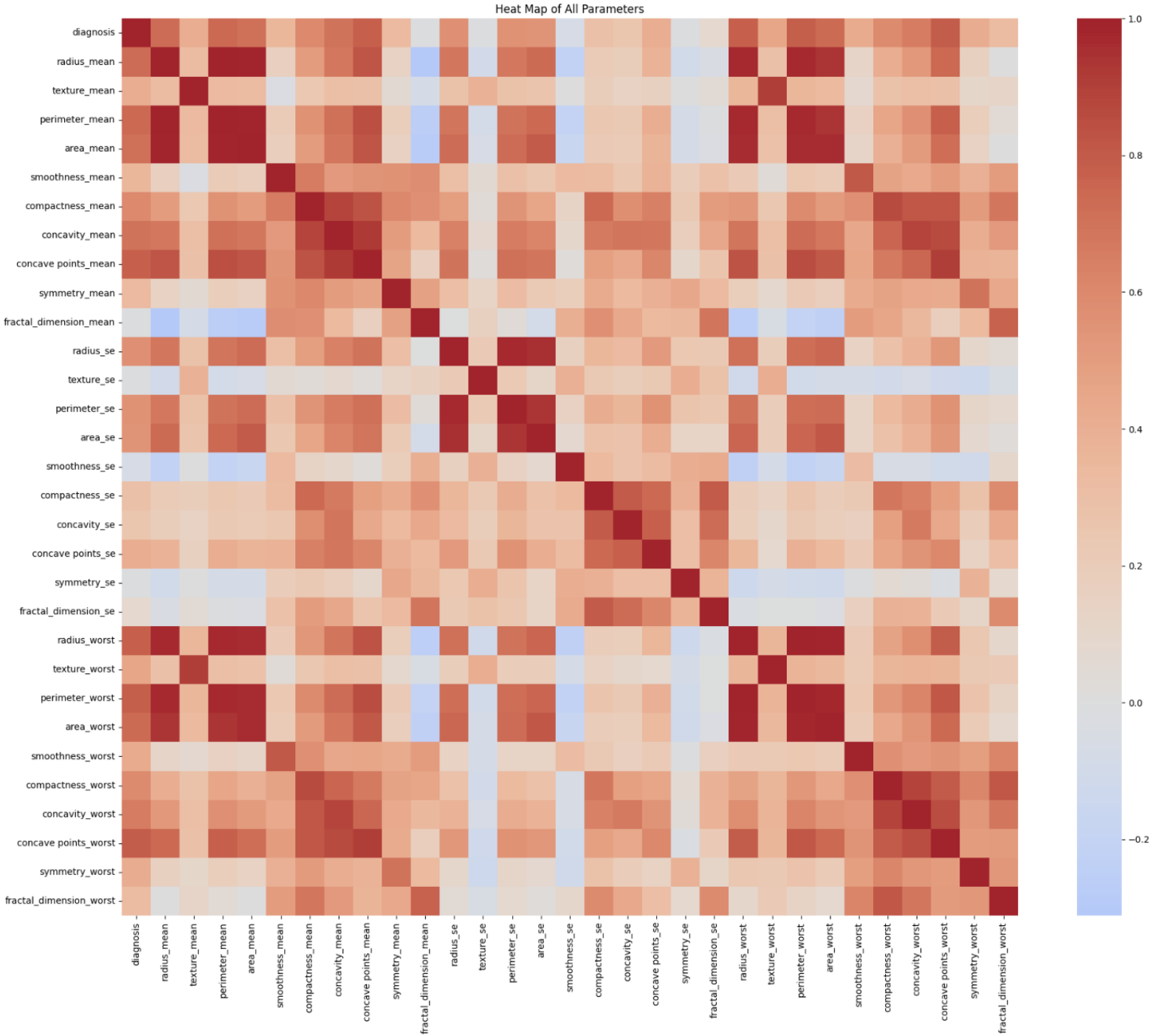
- Removed the ID column as it is nonpredictive
- Checked for missing values. There were none!

Exploratory Data Analysis

Correlations with Malignancy

```
Top correlations with malignancy:
concave points_worst    0.793566
perimeter_worst         0.782914
concave points_mean     0.776614
radius_worst            0.776454
perimeter_mean          0.742636
area_worst              0.733825
radius_mean             0.730029
area_mean               0.708984
concavity_mean          0.696360
concavity_worst         0.659610
compactness_mean        0.596534
compactness_worst       0.590998
radius_se               0.567134
perimeter_se            0.556141
area_se                 0.548236
Name: diagnosis_num, dtype: float64
```

[31 rows x 31 columns]

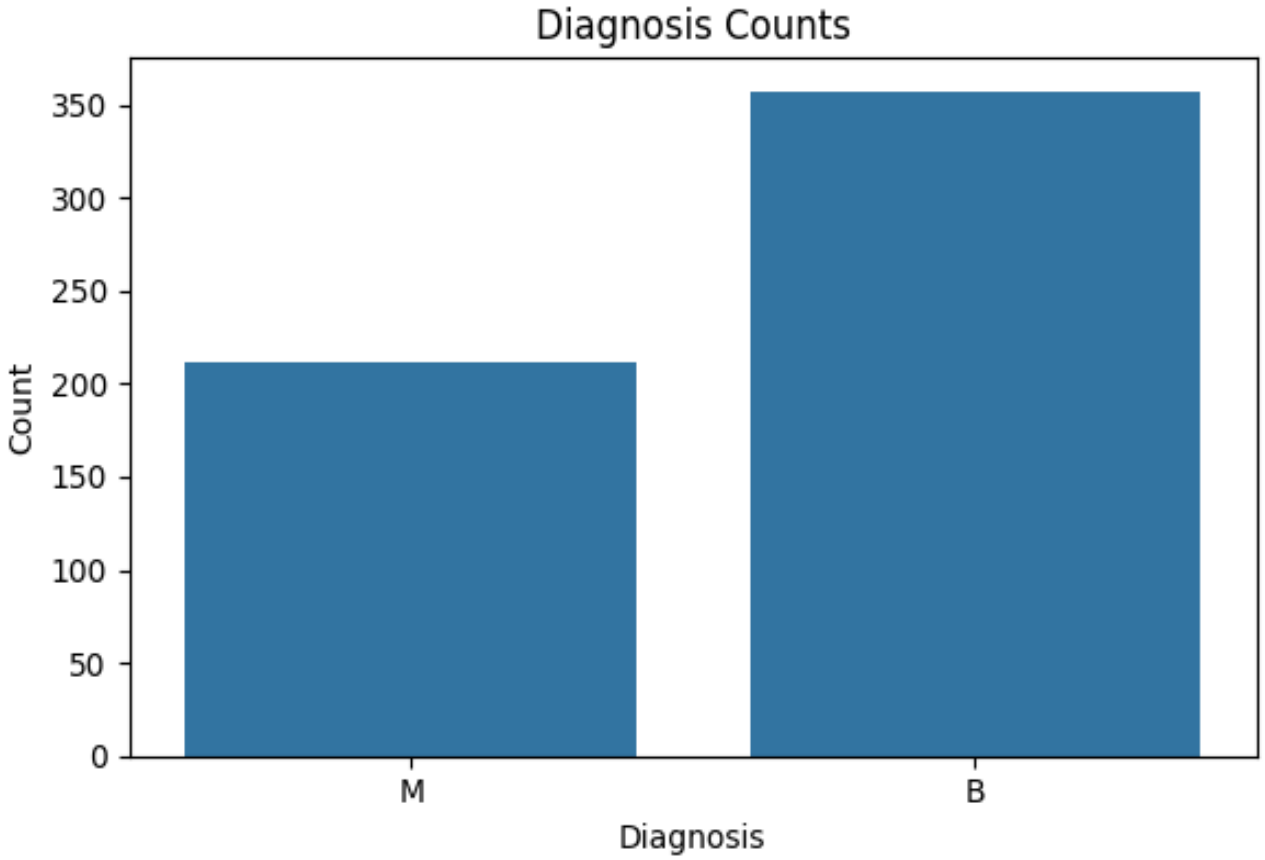


Heat Map- All Parameters

- **Many predictors are highly collinear.**
  - **Ex. Different methods of measurement of the same parameters (cm vs in)**
- Diagnosis has strong positive correlations with size and boundary-irregularity features.
- Using every variable may cause redundancy, making lasso-style step useful.

Class Distribution

- The dataset is not perfectly balanced
- We addressed this in two main ways. First, we used a stratified train-test split to preserve the class proportions in both the training and testing sets. Second, we evaluated the model using AUC and the confusion matrix instead of relying only on accuracy



Train test split (used for both models)

- Split the existing dataset into two datasets by randomly choosing values.
- 80 percent of the identified tumors were for testing; 20 percent were for training.
- The split was stratified to preserve the proportion of benign and malignant cases in both sets. This ensures that the model is evaluated fairly and reduces the risk of biased results.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

Model 1: Binary Lasso (L1 regularization) Based

Lasso Feature Selection (Model 0 L1 Logistic Regression)

- Shrinks weaker coefficients towards zero and keeps the more important predictors.
- An interesting observation is almost all of the shrunk predictors are summary statistics to their extreme counterparts. For example, area_worst was one of the strongest predictors but area_mean was shrunk to 0.

```
area_se                    0.000000
texture_mean               0.000000
perimeter_mean             0.000000
area_mean                  0.000000
smoothness_mean            0.000000
compactness_mean           0.000000
radius_mean                0.000000
perimeter_se               0.000000
symmetry_mean              0.000000
concavity_se               0.000000
symmetry_se                0.000000
concave points_se          0.000000
compactness_worst          0.000000
fractal_dimension_worst    0.000000
```

```
Selected non-zero features:
radius_se                  2.706052
area_worst                 2.434966
radius_worst               2.007194
texture_worst              1.778727
concave points_worst       1.251020
concavity_worst            1.108103
concave points_mean        1.097640
compactness_se            -0.875740
symmetry_worst             0.554144
smoothness_worst           0.547826
texture_se                -0.386619
smoothness_se              0.318290
fractal_dimension_se      -0.234605
fractal_dimension_mean    -0.123714
concavity_mean             0.114773
perimeter_worst            0.044636
```

1. *Start with all tumor predictors*
2. *Standardize them*
3. *Fit a logistic regression with an L1 penalty*
4. *The penalty shrinks less useful coefficients to 0*
5. *The remaining nonzero coefficients indicate the more important predictors*

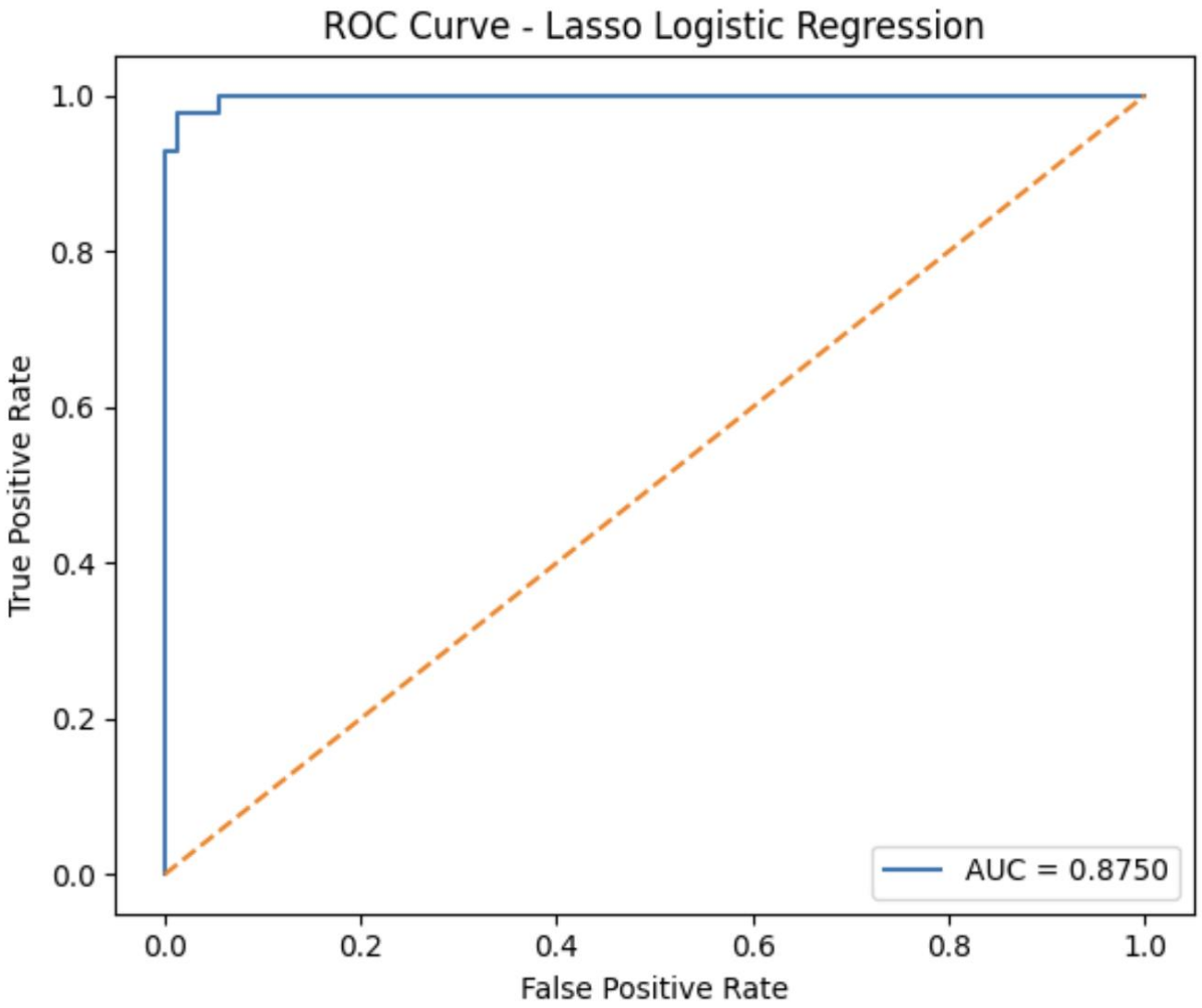
```
("model", LogisticRegression(
    penalty="l1",
```

Lasso Feature Selection cont.

- Since the outcome was binary, we applied logistic regression with an L1 penalty rather than ordinary linear lasso regression. The L1 penalty shrinks weak coefficients toward zero, so the predictors with nonzero coefficients were treated as the most important tumor features.
- Suggests tumor size, irregularity, and concavity are important in separating malignant from benign.
- Not good enough, many malignant tumors were predicted to be benign.

Accuracy = 0.8421

```
Confusion Matrix:
[[54 18]
 [ 0 42]]
```



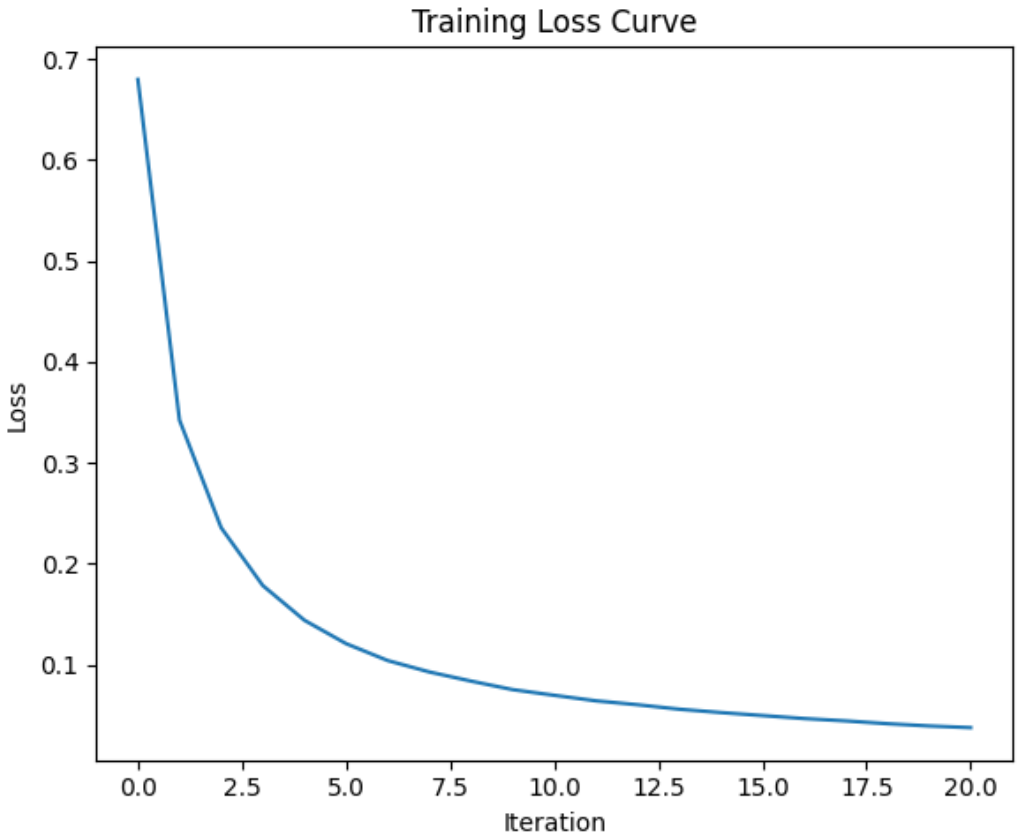
Model 2: Neural Network (FNN) Based

Neural network

- Input layer: each tumor is a vector x
- Hidden layer 1: 32 neurons; Hidden layer 2: 16 neurons
- Each neuron computes h(x)=ϕ * (xT *w + b) where w = weights, b = bias, ϕ = activation function
- For the activation function we used ReLU (rectified linear) in hidden layers

```
model = MLPClassifier(
    hidden_layer_sizes=(32, 16),
    activation="relu",
```

- We used an activation function since they factor in nonlinear relationships between predictors
- Output layer: is one single output node that represents probability of malignant tumor



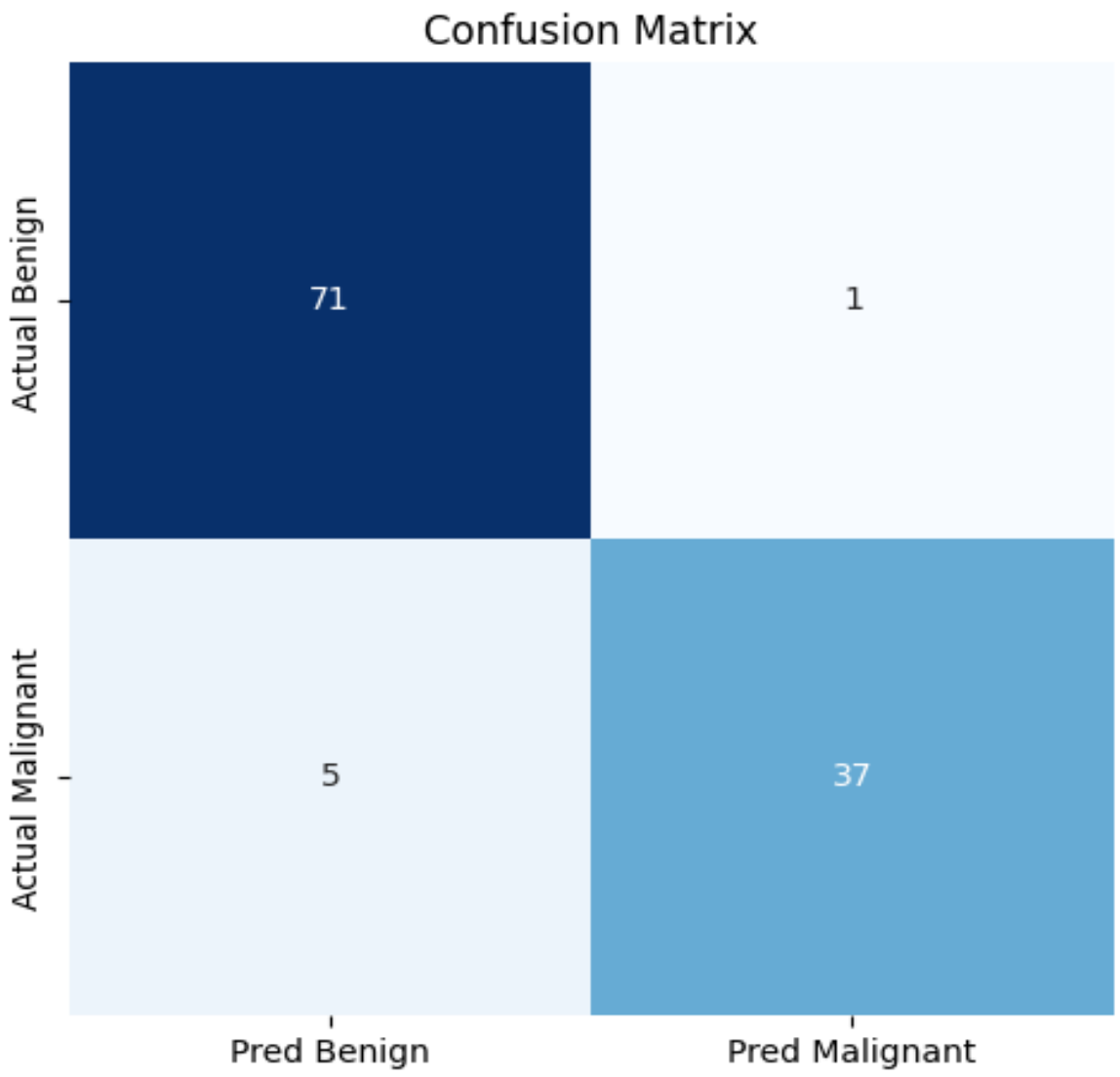
Training Loss Curve

- Loss decreases steadily which means the model is appropriately learning patterns from the data.
- Loss stabilizes meaning the training is complete.

Confusion Matrix & Classification report

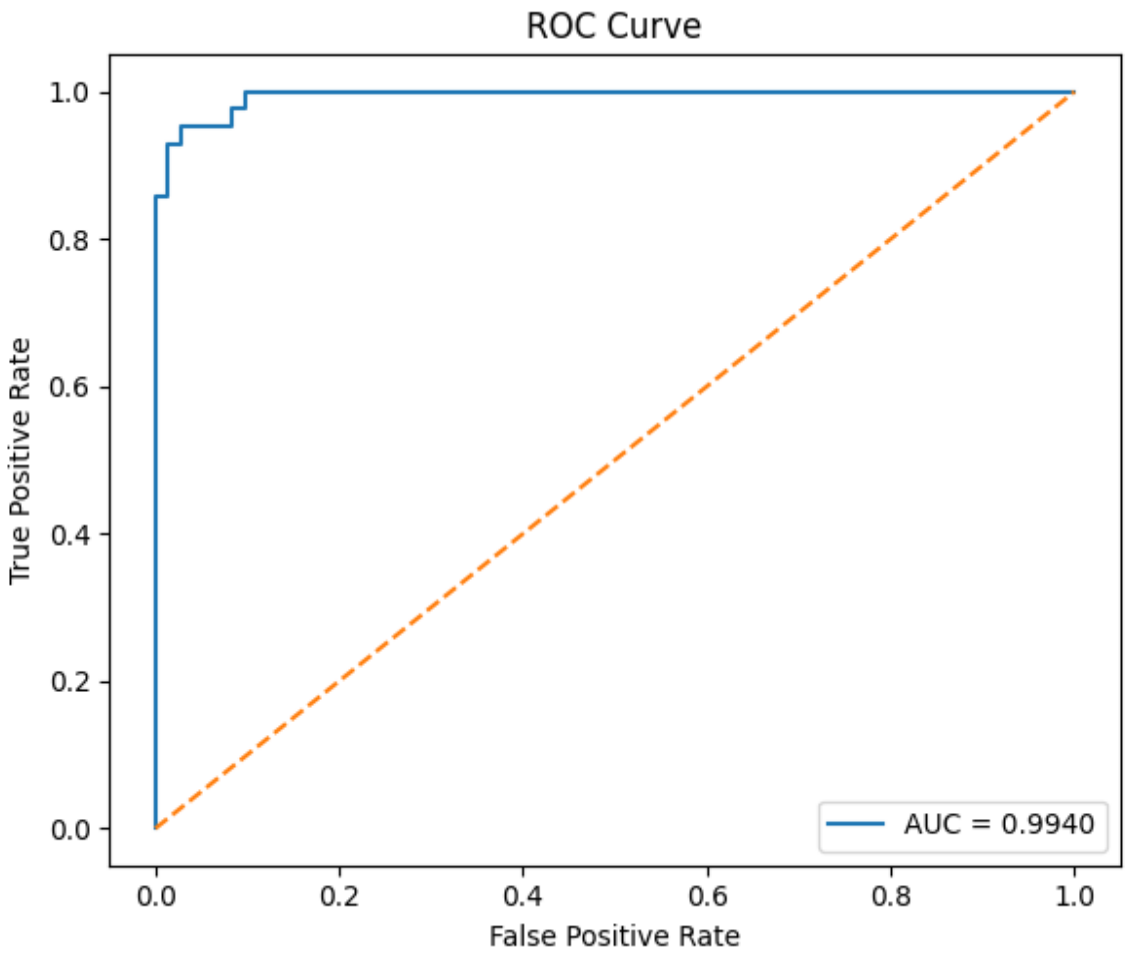
- Excellent separation of benign and malignant tumors.
- Correctly identified almost all benign.
- Missed small amount of malignant.

```
Classification Report:
              precision    recall  f1-score   support

      Benign       0.93      0.99      0.96        72
   Malignant       0.97      0.88      0.93        42

    accuracy                           0.95       114
   macro avg       0.95      0.93      0.94       114
weighted avg       0.95      0.95      0.95       114
```



ROC Curve

- AUC ~0.994, excellent classification of malignant vs benign
- The model has high recall, identifying most malignant tumors, with low false positives
- Medical diagnosis of any FP or FN can be dangerous; a robust model is extremely important.



Conclusions

- From a data scientists lens, we made an extremely accurate model that was able to classify malignant vs benign tumors for breast cancer from the sample parameters utilizing lasso style logistic regression and a neural network.
- Through scaling the most important parameters, from a qualitative lens the most important factors of determining a malignant breast cancer tumor were the predictors that dealt with extreme values of area, radius, and concavity.
- From a medical lens, while the model is highly accurate, just one wrong prediction could spell disaster which is why it is important to continue to imporve cancer detection models:

Improvements/Disscussion

- Our models could be improved in a few ways:
  - Collinearity was a constant problem throughout model training; this could be addressed more thoroughly through further correlation analysis.
  - The errors for this dataset are standardized since it was gathered from the same origin (medical office). Other medical office's tools may have different margins for error therefore pushing the margins for the prediction classifies which mayh result in a more inaccurate model if this model was used on a similar dataset.
  - More emphasis on identifying positives while sacrificing negative accuracy may be more important from a medical lens as further testing can always validate a False negative, if a model ends up predicting a false positive, that person faces a way greater risk as likely no further testing will be conducted.

References

- https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data
- Shockney, L. (2023). Breast Cancer Facts & Statistics. [online] National Breast Cancer Foundation. Available at: https://www.nationalbreastcancer.org/breast-cancer-facts/

thank you